



Assessment: Image Optimization

INTRODUCTION

In this project, you will take an existing image gallery, optimize its images, and audit it for optimal performance and page experience.

Working with the provided starting files, you will audit the gallery in its current state, take note of what needs to be corrected and optimized and apply those changes. Once complete, audit the site to measure the difference.

You will apply techniques for choosing the right next-generation formats, sizing and serving inline, background, and high-definition images, implementing SVG sprites, applying lazy loading correctly, and optimizing based on the critical rendering path. Your Lighthouse scores and written rationale will provide evidence that your decisions were deliberate and grounded in what you learned.

REQUIREMENTS

1. Image Format & Compression
 - Convert raster images to an appropriate next-generation format
 - Ensure file sizes are meaningfully reduced without visible quality loss and sized for their intended display dimensions.
2. Responsive Background Images
 - Export images sized for small, medium and large viewports.
 - Use the defined CSS media queries/breakpoints in the starter file CSS to serve the appropriately sized images.
3. High-Definition (2x) Images
 - Identify which images benefit from 2x version.
 - Export a 2x copy of each identified image and apply it using the defined queries.
4. SVG Sprites
 - Build an SVG sprite file containing all icons used in the gallery UI.
 - All icons need to be accessible to screen readers.
5. Lazy Loading
 - Images load at the appropriate time relative to their position on the page.
 - The most critical image on the page loads with the highest priority.
6. Critical Rendering Path (CRP)
 - Non-critical resources do not delay the initial render.
 - Critical assets are appropriately prioritized.
 - Connections to any external resources are established as early as possible.
7. Lighthouse screenshots before and after scores
 - A before Lighthouse screenshot taken in Chrome DevTools on mobile throttling settings against the unmodified starter files.



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- An after Lighthouse screenshot taken under the same conditions against your completed submission.

8. Short rationale

RATIONALE & LIGHTHOUSE REQUIREMENTS

Include a single-spaced ½ - 1 page (minimum 300 words) written rationale:

- Which format choices you made for which images, and why those formats were appropriate for that content.
- How your responsive background image sizing and 2× decisions were informed by the likely viewing conditions.
- Where and why, you applied (or deliberately did not apply) lazy loading, with reference to the LCP image.
- Which rendering blocking resources were identified? What changes were made to address them and how did those changes reflect in the Lighthouse scores?
- What your Lighthouse before/after scores show, and what you believe is responsible for the change.

Note: the rationale should be checked for spelling and grammatical errors and use a line-height of 1.15 to 1.5 maximum.

SUBMISSION

You must submit the following to Brightspace by the due date and time:

1. Your GitHub Link to your completed **Image Optimization** project folder that was forked from the original repository. This must include your full HTML, CSS, all images and SVGS assets in their final optimized states.
2. In an assets folder placed within your project folder,
 - a. A before and after Lighthouse screenshots taken in Chrome DevTools on mobile throttling settings, named **lighthouse-before.jpeg/png** and **lighthouse-after.jpeg/png**, respectively.
 - b. Your final design rationale file, named **rationale.pdf**.

Late submissions will not be accepted. See Digital Design's Assessment Extension Guidelines.

ACADEMIC INTEGRITY

Use of Generative AI Tools



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For this assessment, the use of any Generative Artificial Intelligence (AI) tools is not permitted. This assessment is intended to measure your ability to generate your own work. Your unique comprehension and interpretation of the course content must be demonstrated in order to be successful. Using prohibited resources will be considered academic misconduct and addressed following SR 1.3 Academic Integrity Procedure.

Incremental Development

You must demonstrate incremental development of your solution. This means you must begin work as soon as possible and commit often to the assignment repository. Each commit must demonstrate functional improvements to the solution. Just like in a professional environment, leaving a task untouched until right before the deadline is not acceptable — start early and make consistent progress.

Ensure all required code comments have been included.

Push your completed assessment to GitHub repository (this repo should be forked from the original) and the link submitted to Brightspace before the due date and time. Late pushes will be counted as late submissions and will not be accepted unless prior arrangements have been made. See Digital Design's Assessment Extension Guideline.